

Create A GUI Program For Raspberry Pi And Arduino

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This article describes the procedure to set up Arduino as a slave device, which gets all control signals from Raspberry Pi running a Python program. The program presented here provides simple steps to interface Raspberry Pi and Arduino. Arduino as a slave can also be used to control electrical devices including LEDs and servos.

Raspberry Pi, a credit-card-sized computer, has a good amount of computing power but lacks the ability to connect to field devices. This is because it is a 3.3V device and has no analogue channels.

Arduino, on the other hand, is a 5V device that can connect with field sensors, and has many analogue, PWM and digital I/O channels. This article attempts to bring out the best of both the worlds by using computing power of both Arduino and Raspberry Pi that can achieve almost anything.

Tkinter interface is used to build a graphical user interface (GUI) to control an I/O pin of Arduino. (Note: Raspberry Pi 4 and Arduino Uno R3 boards were used during testing at EFY Labs.)

Hardware setup of the project is shown in Fig. 1, and block diagram of the project is shown in Fig. 2.

Software development for this article is divided into three parts, as follows:

1. Installing Nanpy libraries on Raspberry Pi operating system
2. Installing Nanpy libraries on Arduino
3. Writing a GUI with Tkinter to control Arduino onboard LED

Installation of Nanpy on Raspberry Pi

The steps for installing Nanpy on Raspberry Pi are given below.

1. Power on Raspberry Pi and connect it to the Internet using Wi-Fi or Ethernet. Open Raspberry Pi terminal,

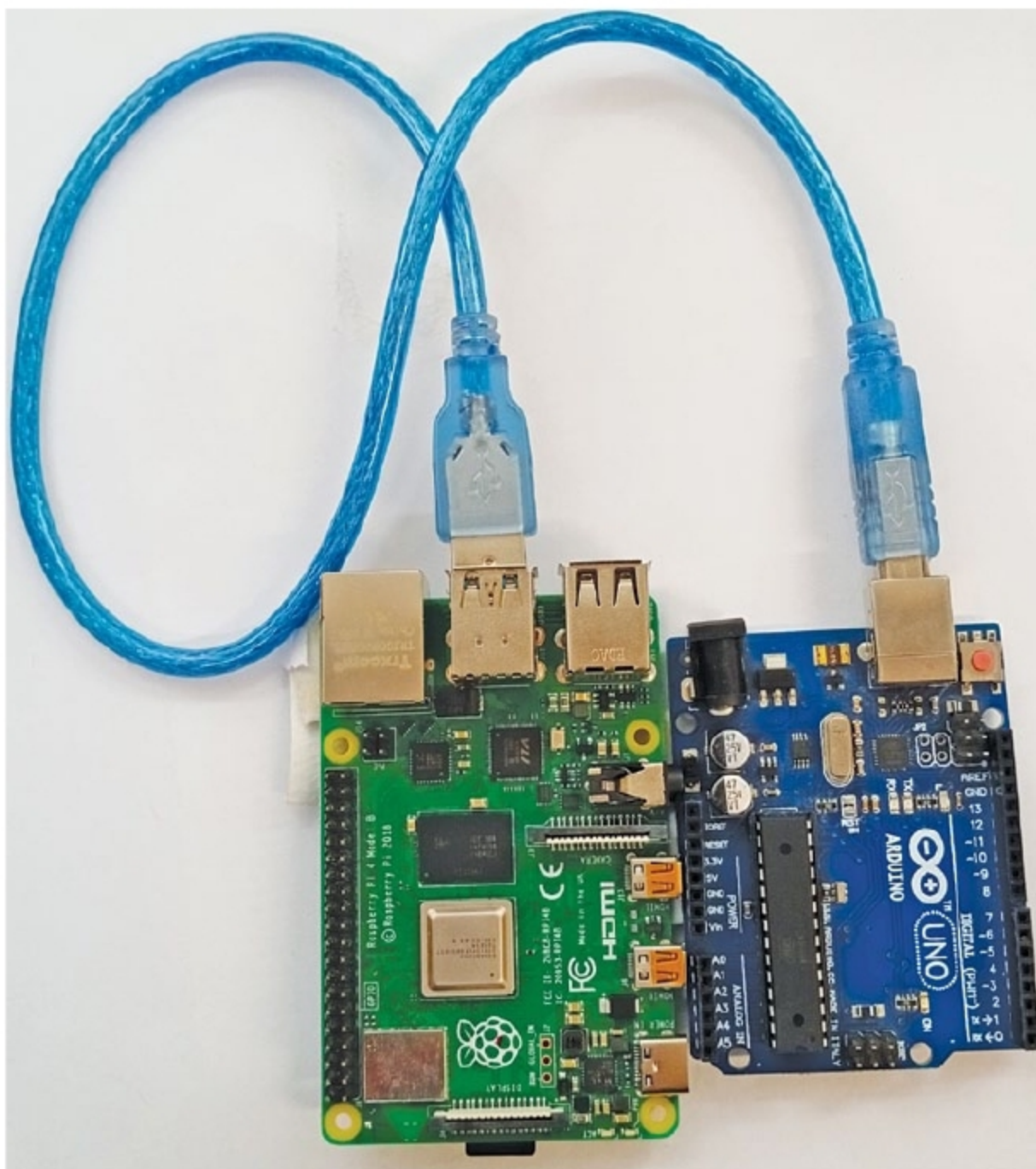


Fig. 1: Project hardware setup

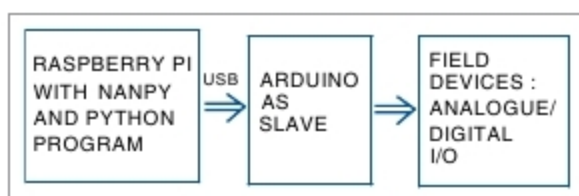


Fig. 2 Block diagram of the project

and the prompt will open as given below.

```
pi@raspberrypi - $
```

2. Enter the following command to install all updates on Raspberry Pi and have all packages up to date:

```
$ sudo apt-get update
```

3. Install Arduino IDE on Rasp-

berry Pi, as given below.

```
$ sudo apt-get install Arduino
```

4. Go to Downloads folder using cd command, as given below.

```
$ cd Downloads
```

Downloads folder may have other contents. It is a good idea to clear all contents from the folder.

When you navigate to Downloads folder, the prompt will be as shown below.

```
$ pi@raspberrypi ~/Downloads/ $
```

5. Enter the following command to install nanpy folder from Github in Raspberry Pi Downloads folder: